Fahimeh Orvati Nia

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Education

Texas A&M University - College Station, TX Jan 2025 - Present
Ph.D. in Computer Engineering (In Progress)
Research Areas: AI, Computer Vision, Agricultural Imaging, Graph Neural Networks
Affiliated with Texas A&M AgriLife Research Institute

University of Notre Dame - Notre Dame, IN Aug 2023 - Jan 2025
M.Sc. in Electrical Engineering (GPA: 3.74)
Thesis: Time-series anomaly detection using GANs and pedestrian intention prediction with GNNs

Amirkabir University of Technology - Tehran, Iran Sep 2016 - Jan 2020
B.Sc. in Electrical Engineering (Control)
Developed a Zigbee-based intelligent HVAC system with enhanced security features.

Work Experience

Texas A&M AgriLife Research - College Station, TX Jan 2025 - Present
Research Assistant

- Computer vision and AI-based analysis of agricultural multispectral imagery.
- Plant growth analysis and phenotyping using deep learning.
- Integrating multispectral imaging with low-field MRI for comprehensive crop analysis.

University of Notre Dame - Notre Dame, IN Fall 2023 - Spring 2024
Teaching Assistant

- Courses: Introduction to Electrical Engineering; Signals and Systems.
- Developed course materials and supported students with assignments and projects.

University of Tehran - Tehran, Iran Jan 2022 - Jun 2023
Research Assistant

- Machine learning for classification and clustering of musical instruments.

NOICT Company - Tehran, Iran Jul 2021 - Oct 2021
IoT Developer

- Wireless communication between IoT devices using the Zigbee protocol.

Amirkabir University of Technology - Tehran, Iran Sep 2020
Teaching Assistant

- Guided students in designing control systems using MATLAB/Simulink.

Skills

Languages & APIs: Python, C/C++, MATLAB
ML & CV: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, CNNs, GNNs, ST-GNNs, ViT, GANs, AEs, SSL
Developer Tools: Git, Docker
Embedded Systems: STM32, Raspberry Pi, Arduino

Papers

- **Fahimeh Orvati Nia**, Hai Lin. *Traffic-aware pedestrian intention prediction.* American Control Conference (ACC), 2025. arXiv:2507.12433.
  Funding: National Science Foundation (CNS-1830335, IIS-2007949).
- **Fahimeh Orvati Nia**, Joshua Peeples, et al. *Neighborhood Feature Pooling for Remote Sensing Image Classification.* **WACV 2026 Workshop on Earth Observation**. arXiv:2510.25077.
  Funding: Texas A&M AgriLife Research.
- **Fahimeh Orvati Nia**, S. Salehi, J. Peeples. *Evaluating GAN-LSTM for Smart Meter Anomaly Detection in Power Systems.* TPEC 2026. arXiv:2601.09701.
- A. McFarland, L. Rossi, F. Nia, J. Peeples, A. Svyanek. *Integrating Multispectral Imaging and Low-Field*

Magnetic Resonance Imaging for Comprehensive Phenotyping of Horticultural Crops. HORTSCIENCE 60 (7), 2025.

Selected Projects

Traffic-Aware Pedestrian Intention Prediction (TA-STGCN) 2025

- Proposed a traffic-aware spatio-temporal GCN integrating dynamic traffic-signal states and contextual cues for pedestrian intention prediction. Improved accuracy on the PIE dataset by **4.75%** over a baseline.

Neighborhood Feature Pooling for Remote Sensing Classification 2025

- Introduced a neighborhood feature pooling layer to capture local relationships and texture-like similarities across feature dimensions. Demonstrated consistent accuracy gains across datasets and backbones with minimal parameter overhead. Evaluated on multiple remote-sensing benchmarks.).

GAN-LSTM for Smart Meter Anomaly Detection 2024-2026

- Conducted a power-system-oriented evaluation of GAN-LSTM anomaly detection on AMI time-series using the LEAD dataset (406 buildings, 1 year). Standardized preprocessing, windowing, and thresholding; achieved strong detection performance (**F1 = 0.89**) versus common baselines.

Agricultural Imaging and Phenotyping (AgriLife Greenhouse) 2025 - Present

- Developing computer-vision methods for analysis of multispectral imagery for plant growth and phenotyping. Building an end-to-end pipeline integrating segmentation, vegetation indices, texture features, and morphological traits. Integrating multispectral imaging with low-field MRI for comprehensive crop phenotyping.

Zigbee-based Intelligent HVAC System (B.Sc. Project) 2019-2020

- Built an IoT HVAC control system using Zigbee communication with enhanced security features. Implemented remote monitoring/control logic and validated end-to-end wireless connectivity.

Activities and Service

Peer Review: LXAI-NeurIPS 2025; IEEE Robotics and Automation Letters (RA-L); ICRA 2026; ACC 2025; CCTA 2025

Student Mentoring: Chloe Tiley; Sophia Martinez-Badillo; Michael Morse; Amran Kassaye; Nazar Oladepo

Awards and Scholarships

- Top rank in Iranian National Entrance Exam for Master's programs (Awarded).
- National full undergraduate scholarship in Iran.
- Ranked top in Computational Intelligence and Communication Systems courses during B.Sc.
- PlantCV Workshop + NAPPN 2026 Travel Award (2026).